

**OBJECT ORIENTED PROGRAMMING  
SECOND YEAR SECOND SEMESTER  
ACTIVITY #1 - INTRODUCTION TO JAVA**

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BSIT 2-1

3/2/2026  
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OBJECT ORIENTED PROGRAMMING  
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ACTIVITY #1 - VARIABLES

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**PART A - VARIABLE IDENTIFICATION TABLE**

| VARIABLE NAME | DATA TYPE             | VARIABLE TYPE | Stored In                        |
|---------------|-----------------------|---------------|----------------------------------|
| String name;  | String                | Instance      | Heap                             |
| int grade;    | Integer               | Instance      | Heap                             |
| s1            | Student (constructor) | Local         | Stack (Reference), Heap (Object) |
| s2            | Student (constructor) | Local         | Stack (Reference), Heap (Object) |
| s3            | Student (constructor) | Local         | Stack (Reference), Heap (Object) |

**PART B - Explanation.** Please see example below.

| STATEMENT              | What happens?                                                  | Memory Impact                                                            |
|------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------|
| Student s1 = new       | With the use of constructor Student,                           | s1 is stored on a stack and                                              |
| Student ("John", 85);  | it stores the value of "John" and 85 for s1                    | adding object on the heap.                                               |
| Student s2 = new       | With the use of constructor Student, it                        | s2 is stored on a stack and                                              |
| Student ("Maria", 90); | stores the value of "Maria" and 90 for s2                      | adding object on the heap.                                               |
| Student s3 = s2;       | It creates duplicate value of s2 for s3                        | s3 is stored on a stack but<br>referenced on the same obj<br>on s2       |
| s1.grade = 95;         | it changes the value of s1.grade to 95                         | The value of object of s1<br>grade is changed from<br>85 to 95.          |
| s3.grade = "Ana"       | It change the value of s3 <del>grade</del> to<br>name as "Ana" | The value of object on<br>s3 grade is changed from<br>and s2 is affected |

**PART C - OUTPUT**

```

John 85
Ana 90
Ana 90
  
```